

CSA4028

NATURAL LANGUAGE PROCESSING

Introduction

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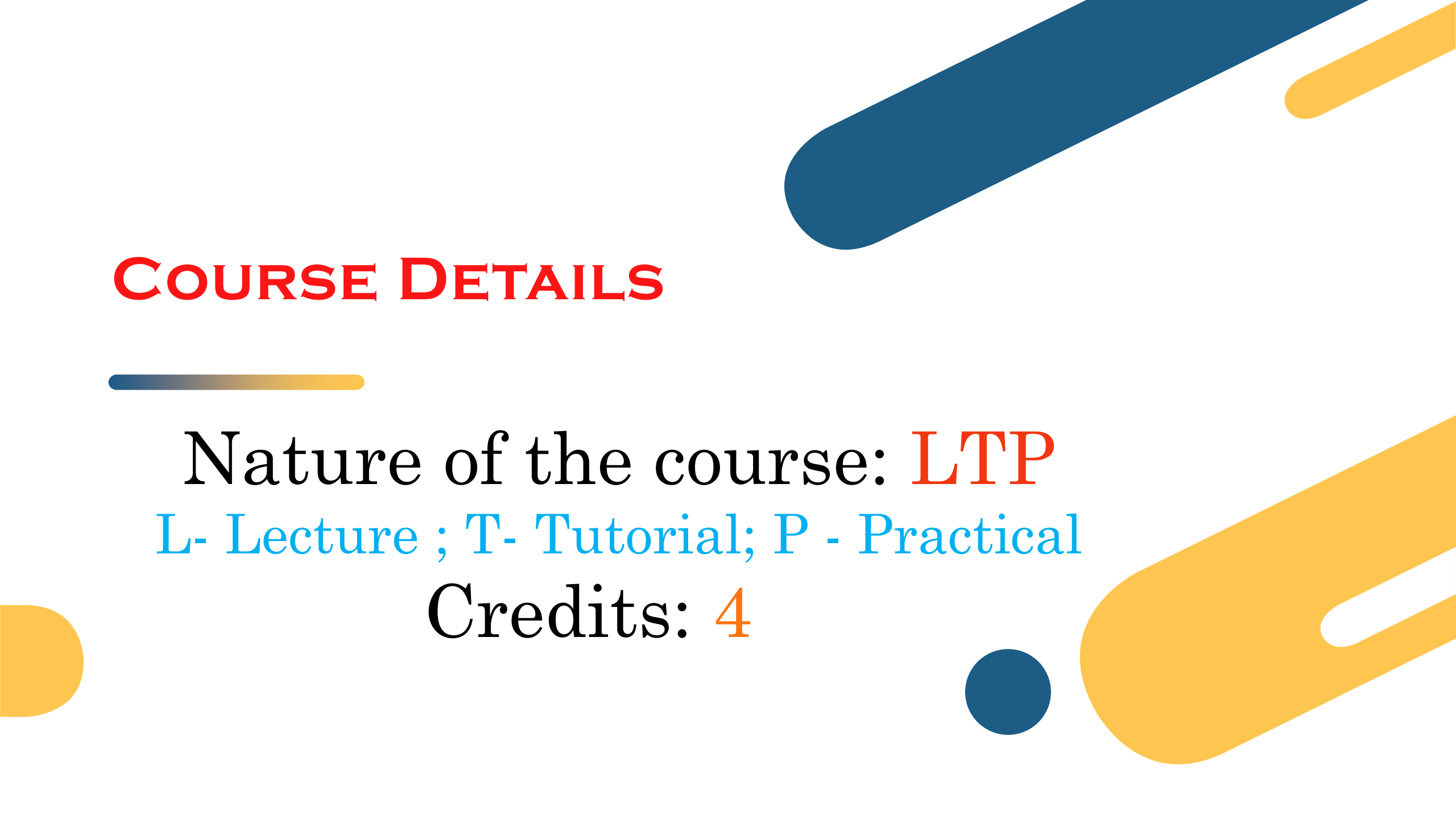
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COURSE DETAILS



Nature of the course: **LTP**
L- Lecture ; T- Tutorial; P - Practical
Credits: 4

UNIT I - INTRODUCTION

Introduction: Knowledge in Speech and Language Processing,
Ambiguity- Models and Algorithms- Language, Thought, and
Understanding- The State of the Art and the Near-Term Future –
Regular Expressions-Basic Regular Expression Patterns-
Disjunction, Grouping, and Precedence- Using an FSA to Recognize
Sheeptalk- Formal Languages.

UNIT II

Morphology And Finite-State Transducers: Inflectional Morphology
- Derivational Morphology- Finite-State Morphological Parsing- The
Lexicon and Morphotactics - Morphological Parsing with Finite-
State Transducers Combining FST Lexicon and Rules- Lexicon-free
FSTs: The Porter Stemmer Human Morphological Processing-
Speech Sounds and Phonetic Transcription- The Phoneme and
Phonological Rules.

UNIT III

Parsing: Tagsets for English - Part of Speech Tagging- Rule- based Part-of-speech Tagging- Stochastic Part-of speech Tagging- Transformation Based Tagging- Chomsky normal form, Context-Free Grammars for English - Context-Free Rules and Trees- The Noun Phrase. The Verb Phrase and Subcategorization- Grammar Equivalence & Normal Form- Finite State & Context-Free Grammars.

UNIT IV

Semantics: Computational Desiderata for Representations-
Meaning Structure of Language- First Order Predicate Calculus-
Elements of FOPC- The Semantics of FOPC- Syntax-Driven
Semantic Analysis- Attachments for a Fragment of English. Regular
Expression: Text Normalization, Decision Tree and Naïve bayes for
Sentiment classification.

UNIT V

Machine Translation And Applications: Basic Issues in Machine Translation- Statistical Translation- Word Alignment- Phrase based Translation- Synchronous Grammars- Applications of Natural Language Processing: Spell Check- Summarization- Language Translation. Case Study: AI Chatbot, Virtual Agent, Humanoids.

Lab Component: List of Experiments

- Preprocessing of text (Tokenization, Filtration, Script Validation, Stop Word Removal, Stemming)
- Morphological Analysis
- N-gram model
- POS tagging
- Virtual Lab on Word Generator
- Building POS Tagger
- Study of IBM Watson, NLTK tools
- Mini Project based on NLP Application

COURSE OUTCOMES



The Students will be able to

- CO1: Explain the concept of language processing and algorithms.
- CO2: Exemplify the Morphology and Finite-State Transducers.
- CO3: Implement the logic of syntax parsing methods.
- CO4: Represent the semantics for language processing.
- CO5: Summarize the approaches for machine translation and applications of NLP.

TEXTBOOKS



1. Daniel Jurafsky and James H. Martin "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech recognition", Prentice Hall, 2nd edition, 2008.
2. Steven Bird, Ewan Klein and Edward Loper "Natural Language Processing with Python", O'Reilly Media; 1 edition, 2009

REFERENCES



1. Roland R. Hausser "Foundations of Computational Linguistics: Human- Computer Communication in Natural Language", Paperback, MIT Press, 2011.
2. Christopher D. Manning and Hinrich Schuetze, "Foundations of Statistical Natural Language Processing" by MIT Press, 1999.
3. Pierre M. Nugues, "An Introduction to Language Processing with Perl and Prolog: An Outline of Theories, Implementation, and Application with Special Consideration of English, French, and German (Cognitive Technologies)", Softcover reprint, 2010.

CONTINUOUS ASSESSMENT

Assessment Components	Number	Weightage
Class Assessment - Tutorial	3 tutorials	5%
Class Assessment – Group Activity	3 activity	5%
Lab Assessment – Continuous Assessment	1 per each experiment	10%
Lab Assessment – Challenging Task	One task	10%
Lab Assessment – Viva	2 Quiz	5%

GROUND RULES FOR THE COURSE

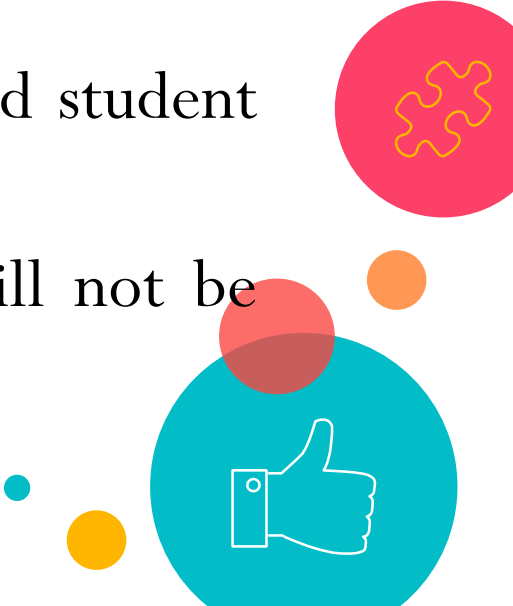
- ❖ The teaching learning process in the course will be interactive and transparent.
- ❖ The students can feel free to ask **any clarifications** regarding the course can **post the question in Google Classroom/** email.
- ❖ Students are expected to be **regular and punctual to the classes**- late comers (more than 5 minutes) can attend the class but will be marked absent for the hour.
- ❖ Students are requested to give their **valuable suggestion** to improve overall teaching – learning process at any point of time.



GROUND RULES FOR THE COURSE

- ❖ Submission of Tutorial and other task related to course should be completed before the deadline.

Plagiarism Policy

- ❖ Students are expected to **avoid plagiarism** in assessments that are planned for the course.
 - ❖ Plagiarism if found any, the students involved (student copied and student helped to copy) has to **resubmit for first time**.
 - ❖ If found repeated for the second time, his / her documents will not be evaluated and zero marks for submission will be awarded.
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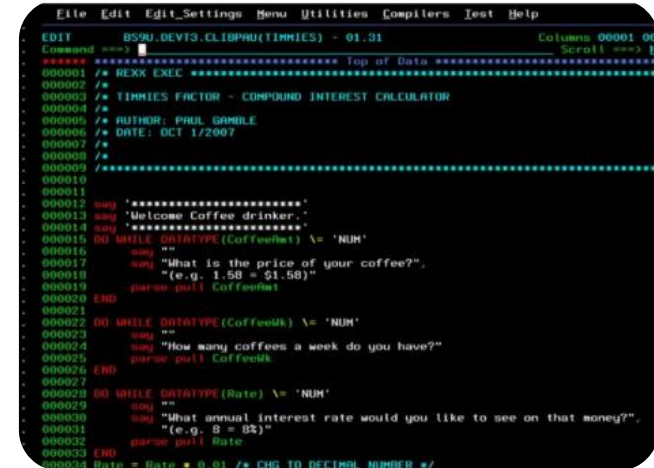
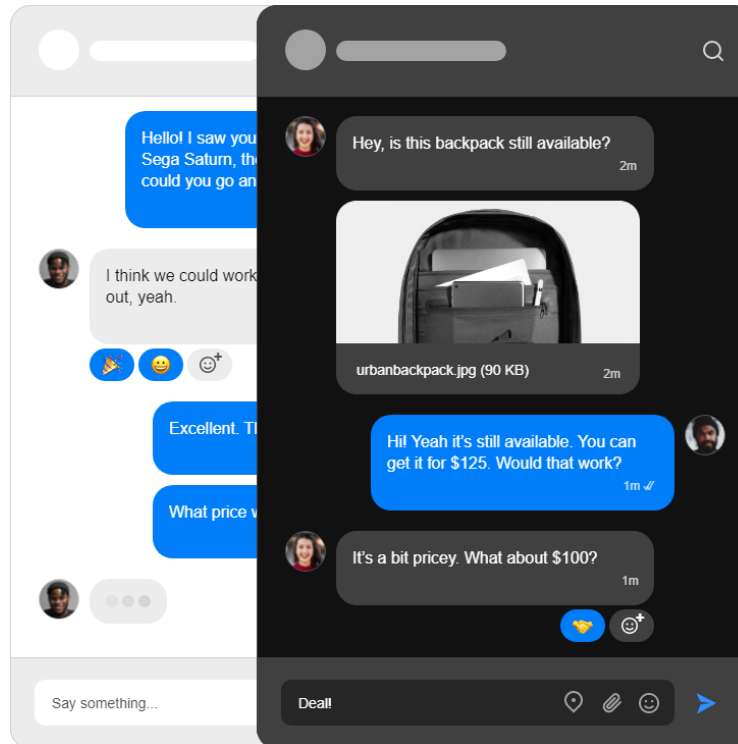
INTRODUCTION TO NLP

COMMUNICATION WITH MACHINES

Today



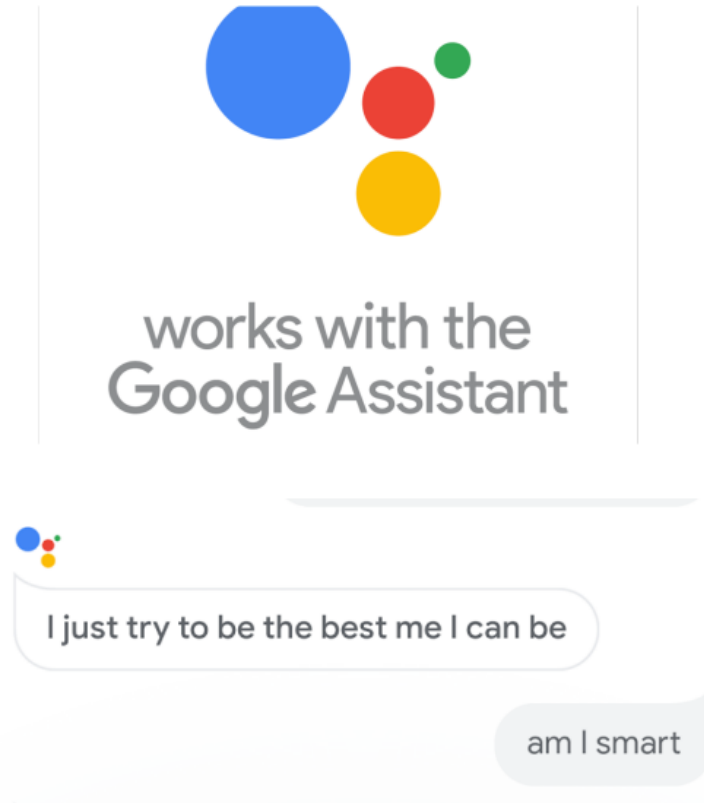
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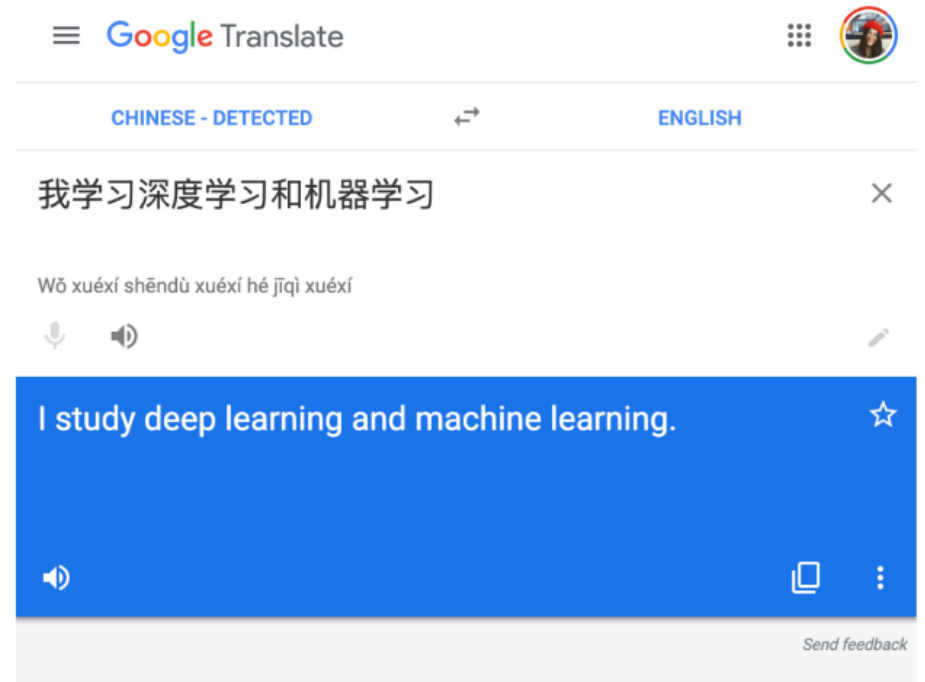
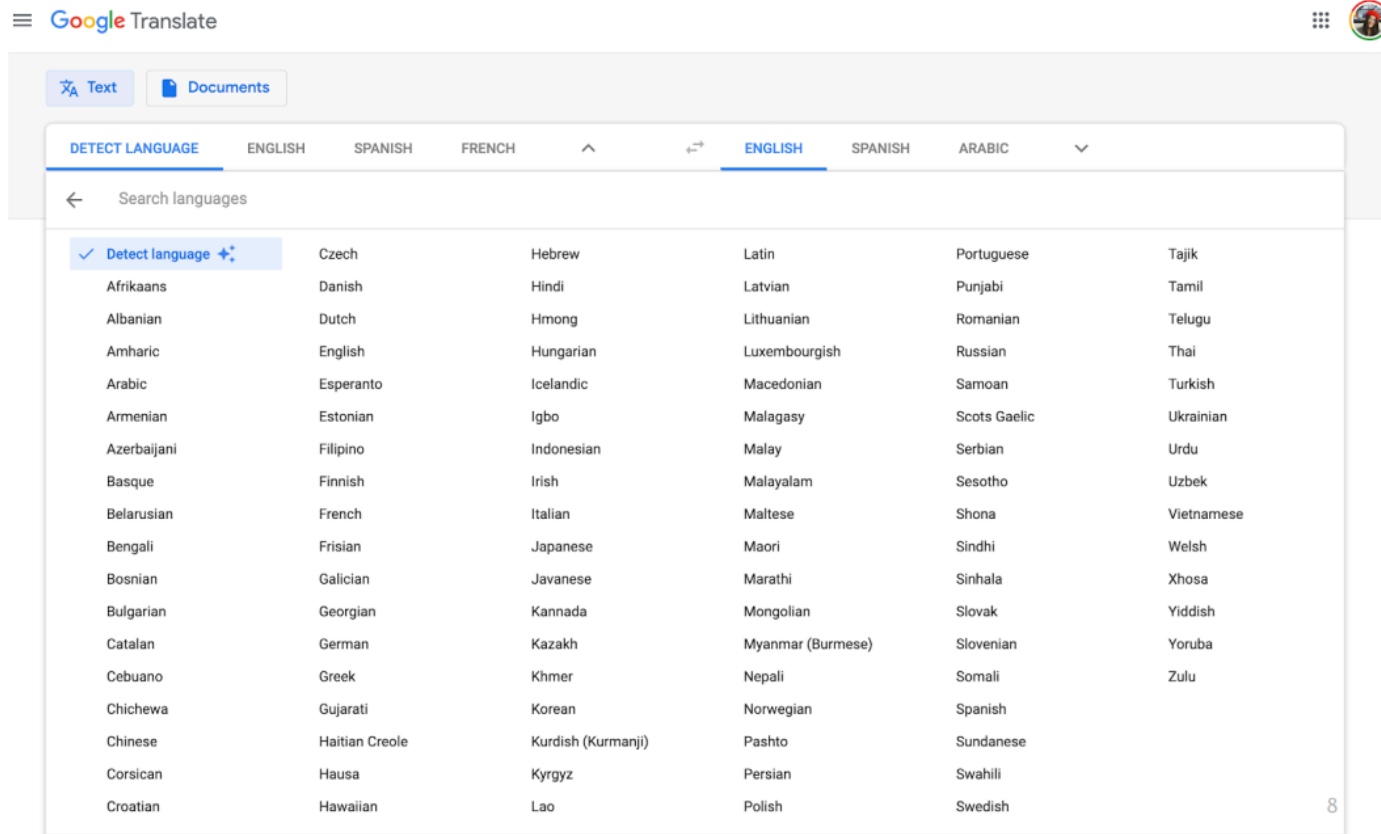
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CONVERSATIONAL AGENTS

- Conversational agents contain:
 - Speech recognition
 - Language analysis
 - Dialogue processing
 - Information retrieval
 - Text to speech



MACHINE TRANSLATION



NATURAL LANGUAGE PROCESSING



Applications

- Machine Translation
- Information Retrieval
- Question Answering
- Dialogue Systems
- Information Extraction
- Summarization
- Sentiment Analysis

Core Technologies

- ¡ Language modeling
- ¡ Part-of-speech tagging
- ¡ Syntactic parsing
- ¡ Named-entity recognition
- ¡ Word sense disambiguation
- ¡ Semantic role labeling
- ¡ ...

APPLICATIONS

Information
Retrieval



Sentiment
Analysis



Information
Extraction



Machine
Translation



**Natural Language
Processing
(NLP)**

Question
Answering

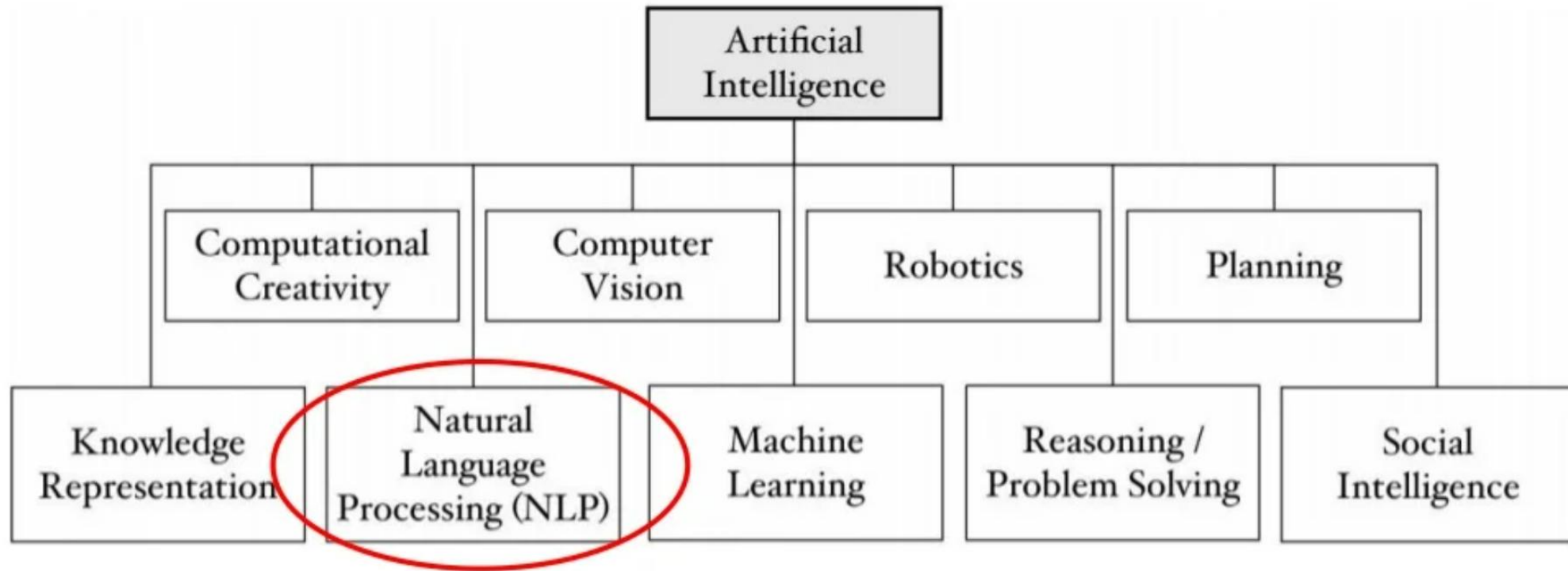


WHAT IS NLP?



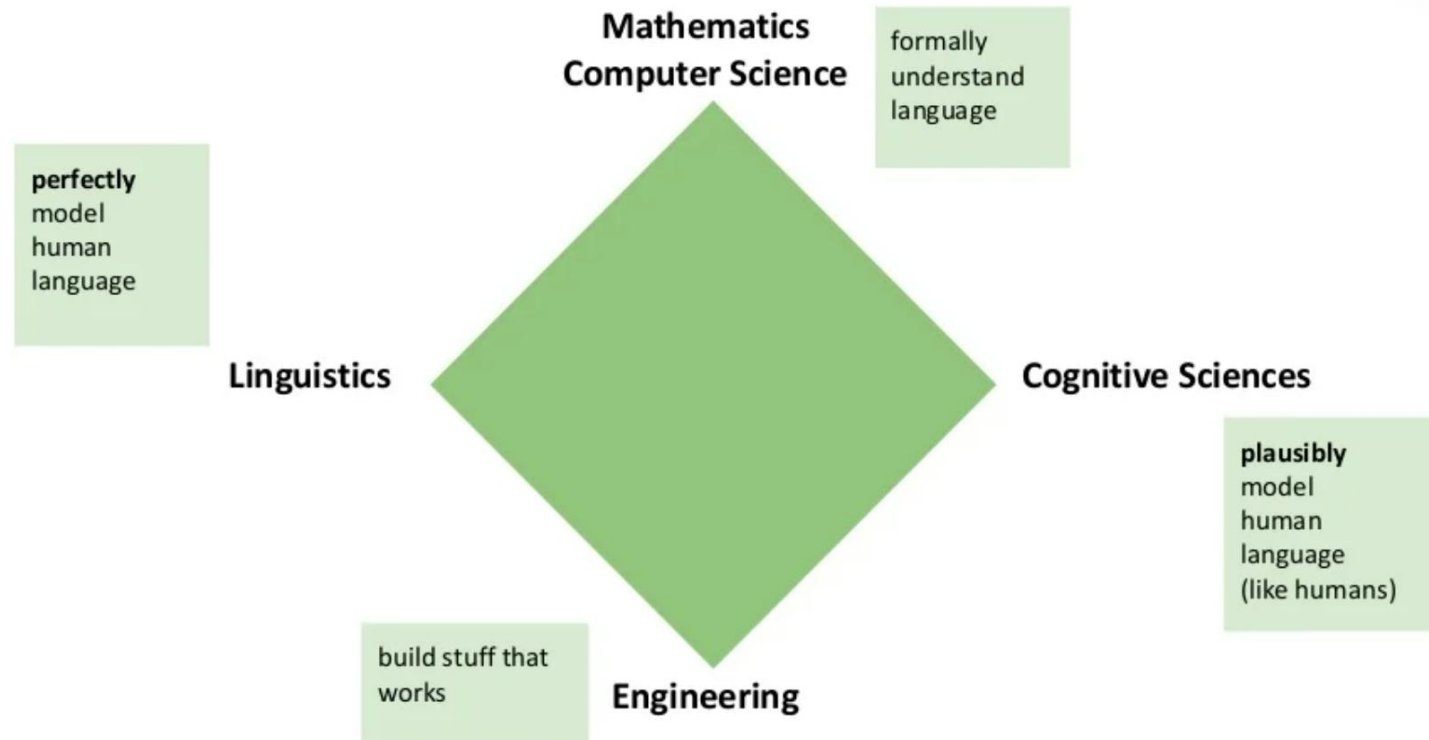
- Natural Language Processing (NLP) is the ability of a computer program to understand human language as it is spoken and written - referred to as **natural language**. It is a component of **Artificial Intelligence (AI)**.
- NLP has existed for more than 50 years and has roots in the field of linguistics. It has a variety of real-world applications in a number of fields, including **medical research, search engines and business intelligence**.

POSITIONING OF NLP IN AI



Picture Taken from: <https://www.slideshare.net/lysander07/02-ise-2021-natural-language-processing-1>

NLP IS A INTERDISCIPLINARY SCIENCE



Picture Taken from: <https://www.slideshare.net/lysander07/02-ise-2021-natural-language-processing-1>



KNOWLEDGE IN SPEECH AND LANGUAGE PROCESSING

CONTD...



- By speech and language processing - computational techniques that process spoken and written human language, as language.
- **Unix wc program**, which is used to count the total number of bytes, words, and lines in a text file.
- ‘wc’ is an ordinary **data processing application**.
- when it is used to **count the words in a file** it requires knowledge about what it means to be a word, and thus becomes a **language processing system**.

CONTD...



- To determine what 'Dave' is saying, HAL must be capable of analyzing an **incoming audio signal** and **recovering the exact sequence of words** Dave used to produce that signal.
 - **In response** - HAL must be able to take a sequence of words and generate an audio signal that Dave can recognize.
 - Both of these tasks require knowledge about **phonetics** and **phonology**.
 - HAL is capable of producing contractions like **I'm and can't**.
- 

CONTD...



- Producing and recognizing these and other variations of individual words
- For example **recognizing that doors** is plural - requires knowledge about **morphology**.

The following variations of his original statement.

- **HAL, the pod bay door is open.**
- **HAL, is the pod bay door open?**

CONTD...



In addition, HAL must use similar structural knowledge to properly string together the words that constitute its response. For example, HAL must know that the following sequence of words will not make sense to Dave, despite the fact that it contains precisely the same set of words as the original.

○ **I'm I do, sorry that afraid Dave I'm can't.**

The knowledge needed to order and group words together comes under the heading of syntax.



CONTD...



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CONTD...



- **knowledge of the meanings** of the component words, the domain of **lexical semantics**, and knowledge of how these components combine to form larger meanings, **compositional semantics**.
- The appropriate use of polite and indirect language comes under the heading of **pragmatics** .

CONTD...



The knowledge of language needed to engage in complex language behavior can be separated **into six distinct categories**.

- **Phonetics and Phonology** – The study of linguistic sounds.
 - **Morphology** – The study of the meaningful components of words.
 - **Syntax** – The study of the structural relationships between words.
 - **Semantics** – The study of meaning.
 - **Pragmatics** – The study of how language is used to accomplish goals.
 - **Discourse** – The study of linguistic units larger than a single utterance.
- 

LEVELS OF NLP



- Level 1 – Speech sound (Phonetics & Phonology)
- Level 2 – Words & their forms (Morphology, Lexicon)
- Level 3 – Structure of sentences (Syntax, Parsing)
- Level 4 – Meaning of sentences (Semantics)
- Level 5 – Meaning in context & for a purpose (Pragmatics)
- Level 6 – Connected sentence processing in a larger body of text (Discourse)

EXAMPLES



- L1 : sound
- L2 : Dog - Dog(s), Dog(ged)
Lady – Lad(ies)

Should we store all forms of words in the lexicon?

- L3 : Ram goes to market (right)
goes Ram to the market (wrong)
- L4 : translation from unstructured to structured representation
go : (event)
agent : Ram
source : ?
destination : market)

EXAMPLES



- L5 : User situation & context

“Is that water?” – the action to be performed is different in a chemistry lab and on a dining table.
- L6 : Backward & forward references – Coreference resolution

“The man went near the dog. It bit him.”

Often co reference & ambiguity go together as in –

“The dog went near the cat. It bit it.”

The background features several abstract geometric shapes. A large, dark blue rounded rectangle is positioned in the top right corner. A smaller, yellow rounded rectangle is located in the bottom right corner. A horizontal bar with a blue-to-yellow gradient is situated in the middle left area. A small dark blue circle is placed near the bottom center. A yellow semi-circle is visible on the far left edge.

AMBIGUITY

AMBIGUITY



- All tasks in speech and language processing can be viewed as resolving ambiguity at one of these levels.
- Consider the spoken sentence **I made her duck**
- Five different meanings this sentence could have (there are more) each of which exemplifies an ambiguity at some level:
 - (1.1) I cooked waterfowl for **her**.
 - (1.2) I **cooked** waterfowl belonging to her.
 - (1.3) I **created** the (plaster?) **duck** she owns.
 - (1.4) I caused her to quickly lower her head or body.
 - (1.5) I waved my magic wand and turned her into undifferentiated waterfowl.

CONTD...



- First, the words **duck and her** are morphologically or syntactically ambiguous in their part of speech.
 - **Duck** can be a verb or a noun
 - **her** can be a dative pronoun or a possessive pronoun
 - Second, the word **make** is semantically ambiguous; it can mean **create or cook**.
 - Finally, the **verb make** is syntactically ambiguous in a different way.
 - Finally, **make** can take a direct object and a verb (1.4), meaning that the object (her) got caused to perform the verbal action (duck)
- 

CONTD...



- **Models and Algorithms** – resolves ambiguity
- For example deciding whether **duck** is a **verb or a noun** can be solved by part of speech tagging.
- Deciding whether **make** means ‘**create**’ or ‘**cook**’ can be solved by word sense disambiguation.
- Deciding whether her and duck are part of the same entity (as in (1.1) or (1.4)) or are different entity (as in (1.2)) can be solved by **probabilistic parsing**.

CONTD...



- Ambiguities that don't arise in this particular example (like whether a given sentence is a statement or a question) will also be resolved, for example by **speech act interpretation**